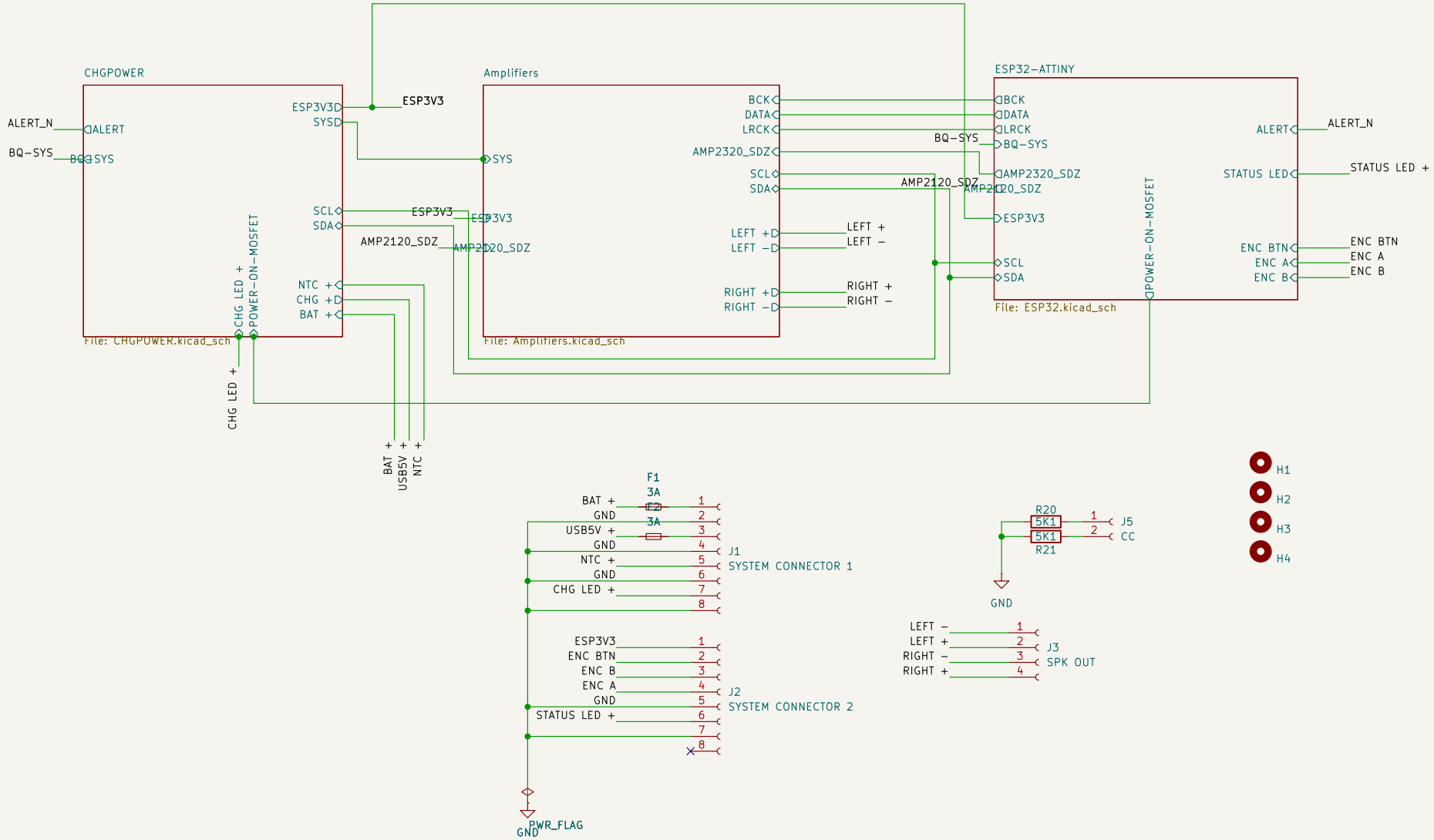
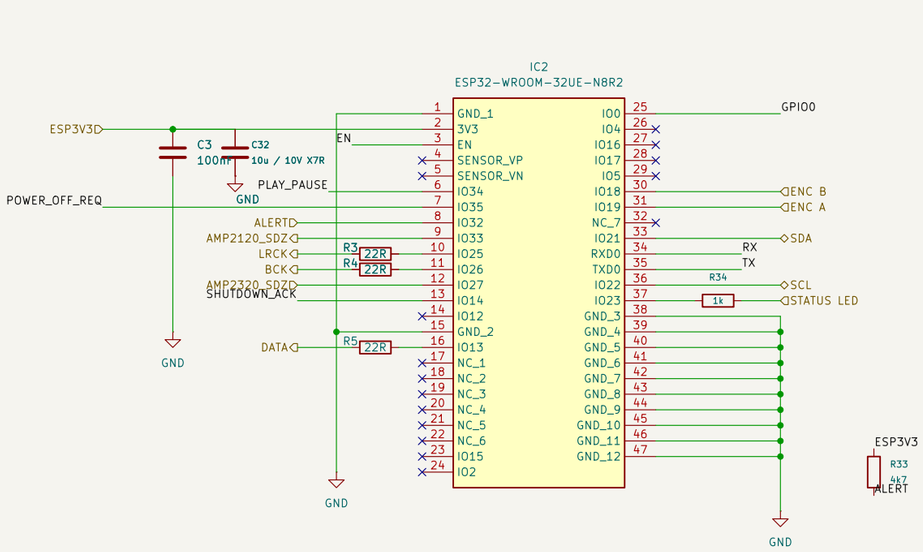
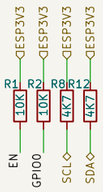
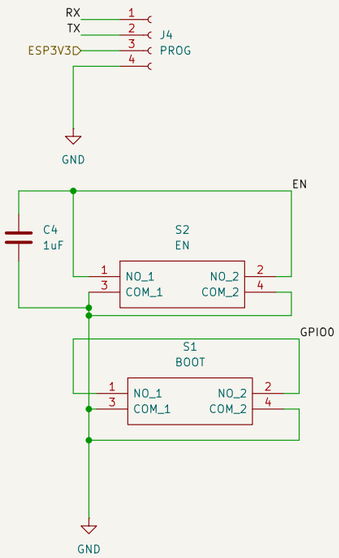
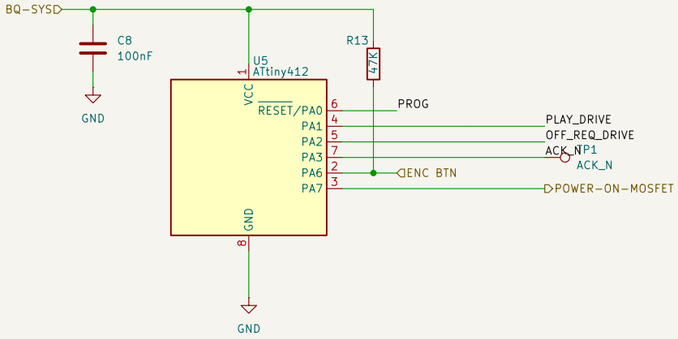


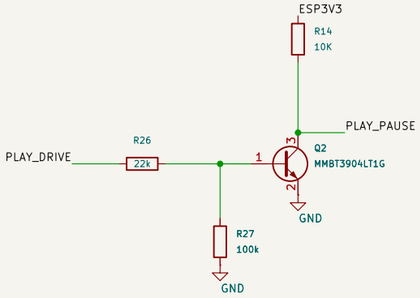
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 1/4

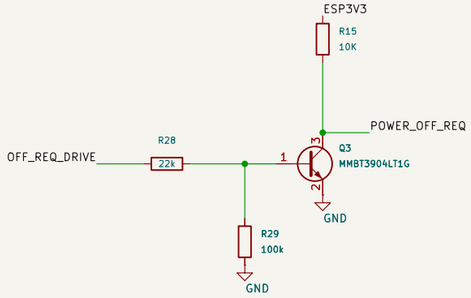


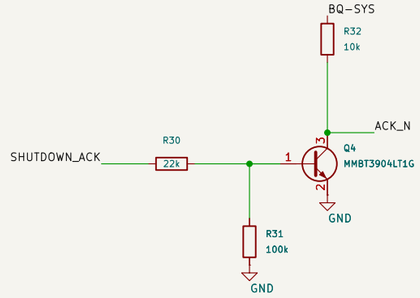




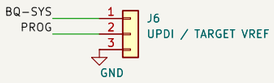






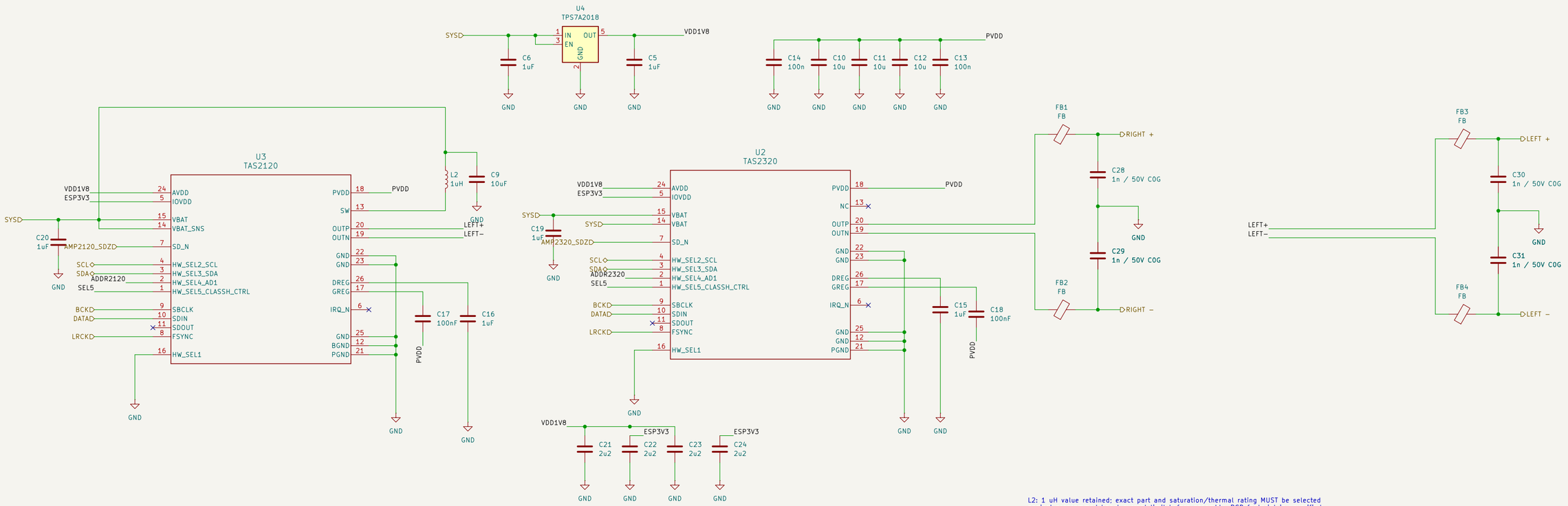


POWER CONTROL (always-on ATtiny): PA1/PA2 HIGH assert active-LOW ESP requests.
ESP GPIO14 HIGH asserts ACK_N LOW at ATtiny PA3. Pull GPIO14 LOW at boot.
Tiny: latch Q1 ON, request shutdown, wait for ACK_N or bounded timeout, then release Q1.
ESP: mute amplifiers / SDZ LOW, finish state writes, then assert ACK.
Firmware changes required; transistor interfaces invert logic and prevent rail backfeed.

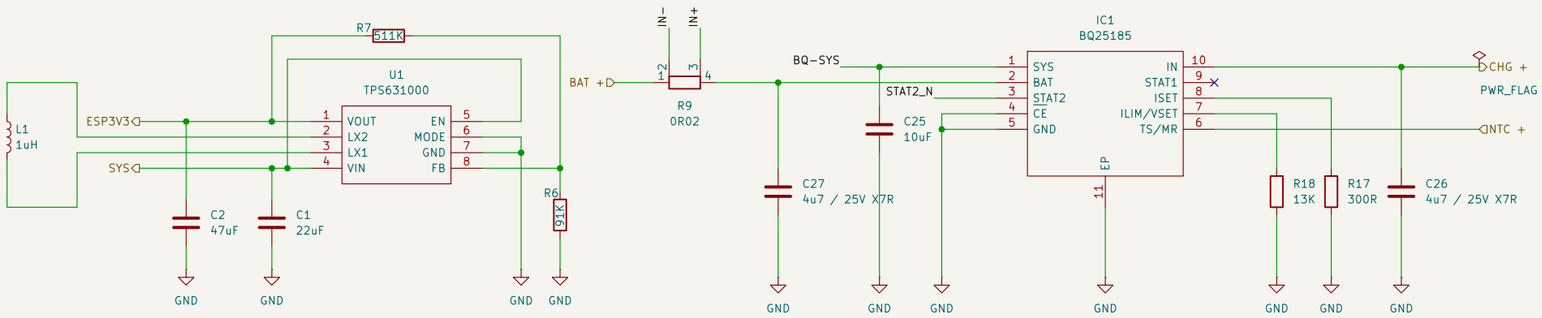


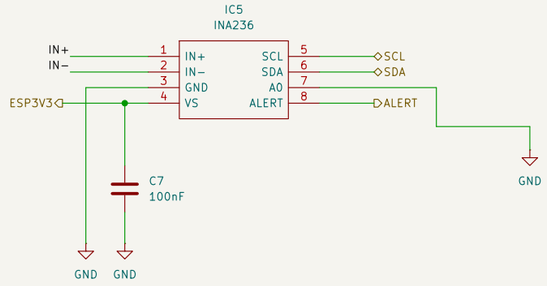
J6: 1 = target VREF (BQ-SYS), 2 = UPDI, 3 = GND.
Use a voltage-compatible programmer; do not back-power this rail.

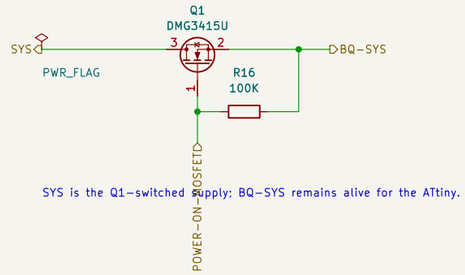
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KiCad E.D.A. 10.0.5		Id: 2/4



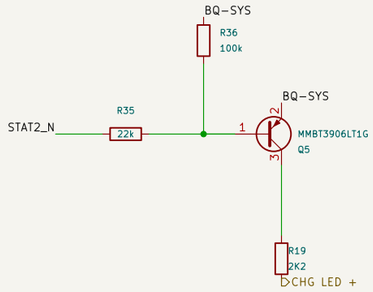
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KiCad E.D.A. 10.0.5	**Rev:** Id: 3/4







SYS is the Q1-switched supply; BQ-SYS remains alive for the ATtiny.



L1: verify exact 1 uH part, Isat above 3.35 A with margin and adequate RMS rating.
R9: verify exact 20 mOhm four-terminal shunt and footprint/Kelvin pad mapping.
Neither unspecified physical part has been substituted.

BQ25185 settings retained: R17 300R => 1 A charge; R18 13k => 1.1 A input limit / 4.2 V.
7 Ah single Li-ion pouch: six-hour safety timer can prevent a full charge from deep discharge.
System load and thermal regulation further reduce charging rate. Charger choice remains a design limitation.
C26/C27: require at least 1 uF effective capacitance. Confirm DC-bias derating.
Use protected cell/pack and specified 10k NTC (beta 3435 K); do not omit cell protection.

Sheet: /CHGPOWER/
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KiCad E.D.A. 10.0.5 | Id: 4/4